DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH, LUCKNOW

EVALUATION SCHEME & SYLLABUS

FOR

B. TECH. THIRD YEAR

ELECTRICAL AND COMPUTER ENGINEERING

Based On

NEP2020

(Effective from the Session: 2024-25)

Electrical and Computer Engineering
B.Tech. V Semester

SEMESTER- V													
S. No.	**Course Code**	**Course Title**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credits**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	BEE501	Power System - I	3	1	0	20	10	30		70		100	4
2	BEE502	Control System	3	1	0	20	10	30		70		100	4
3	BEE503	Electrical Machines-II	3	1	0	20	10	30		70		100	4
4	BEE052/ BEE053/ BEE055/ BEEZ051	Department Elective-I	3	0	0	20	10	30		70		100	3
5	BCS051/ BCS053/ BCS056/ BEEZ052	Department Elective-II	3	0	0	20	10	30		70		100	3
6	BEE551	Power System-I Lab	0	0	2				50		50	100	1
7	BEE552	Control System Lab	0	0	2				50		50	100	1
8	BEE553	Electrical Machines – II Lab	0	0	2				50		50	100	1
9	BEEZ554	Mini Project/Internship **	0	0	2				100			100	2
10	BNC501/ BNC502	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			
		TOTAL	**17**	**3**	**8**							**900**	**23**

Minor Degree/Honors Degree MT-1/HT-1

**The Mini Project or Internship (4 weeks) conducted during summer break after IVth Semester and will be assessed during Vth Semester.

**It is desirable that the students should do their Summer Internship and Minor Project in their Specialization area in line with the B.Tech. program.

Departmental Elective-I

1. BEE052 Sensors and Transducers
2. BEE053 Industrial Automation and Control
3. BEE055 Optimization Techniques
4. BEEZ051 Operating System

Departmental Elective-II

1. BCS051 Statistical Computing
2. BCS053 Computer Graphics
3. BCS056 Application of Soft Computing
4. BEEZ052 Object Oriented Programming with Java

ELECTRICAL AND COMPUTER ENGINEERING

B.Tech. VI Semester

SEMESTER- VI													
S. No.	**Course Code**	**Course Title**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credits**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	BEE602	Microprocessor	3	1	0	20	10	30		70		100	4
2	BEE603	Power Electronics	3	1	0	20	10	30		70		100	4
3	BCS603	Computer Network	3	1	0	20	10	30		70		100	4
4	BCS063/ BCS064/ BEE061/ BEEZ061/ BEEZ062	Department Elective–III	3	0	0	20	10	30		70		100	3
5	BOE06*	Open Elective-I	3	0	0	20	10	30		70		100	3
6	BEE652	Microprocessor Lab	0	0	2				50		50	100	1
7	BEE653	Power Electronics Lab	0	0	2				50		50	100	1
8	BCS653	Computer Network Lab	0	0	2				50		50	100	1
9	BNC601/ BNC602	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			
		Total	**17**	**3**	**6**							**800**	**21**

Minor Degree/Honors Degree MT-1/HT-1

*The Mini Project or internship (4 weeks) will be done during summer break after the 6th Semester and will be assessed during 7th semester.

*It is desirable that the students should do their Summer Internship and Minor Project in their specialization area in line with the B.Tech. program.

Department Elective-III

1. BCS063 Blockchain Architecture Design
2. BCS064 Data compression
3. BEE061 Special Electrical Machines
4. BEEZ061 Web Technology
5. BEEZ062 Theory of Automata and Formal Languages

BEE501: POWER SYSTEM-I

Pre-requisites of the course: Basic Electrical Engineering, Networks Analysis and Synthesis, Electromagnetic Field Theory.

Course Outcome (CO)		Bloom's Knowledge Level (KL)
Upon the completion of the course, the student will be able to:		
CO 1	Describe the working principle and basic components of conventional power plants as well as the other aspects of power generation.	**K2**
CO 2	Recognize elements of power system and their functions, as well as compare the different types of supply systems. Illustrate different types of conductors, transmission lines and various performance parameters of transmission line for short, medium and long transmission line.	**K3**
CO 3	Calculate sag and tension in overhead lines with and without wind and ice loading. Classify different type of insulators, determine potential distribution over a string of insulator, string efficiency and its improvement.	**K3**
CO 4	Compute the inductance and capacitance of single phase, three phase lines with symmetrical and unsymmetrical spacing, Composite conductors-transposition, bundled conductors, and understand the effect of earth on capacitance of transmission lines.	**K3**
CO 5	Elucidate different types of cables and assess the Resistance and capacitance parameters of cables, grading of cables and compare overhead lines and cables.	**K3**

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)
K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create

DETAILED SYLLABUS **3-1-0**

UNIT-I (Power Generation):

Introduction: Basic structure of power system, sources of electric energy: conventional and non-conventional; Layout of Hydro-electric, Thermal and Nuclear power plants, Concept of cogeneration, combined heat and power, and captive power plants, SPV generation at household level. Basic Structure of Smart Grid, Communication and IT components in power system.Load curve, load duration curve, Concept of Connected Load, Maximum Demand, Average load, Demand Factor, Load factor, Diversity Factor, Capacity Factor, Utilization factor, Plant use factor, Installed capacity, Reserves, role of load diversity in power system economy. Load Sharing between Base load and Peak Load

UNIT-II (Transmission & Distribution of Electric Power- I):

Single line diagram of Power system, choice of transmission voltage, One-line diagram, Impedance and reactance diagram, per unit system changing the base of per unit quantities, advantages of per unit system Different kinds of supply system and their comparison.

Configurations of transmission lines: Types of conductors, Bundled Conductors, resistance of line, skin effect, Kelvin's law, Proximity effect, Corona Effect, factors affecting the Corona, Corona Power Loss, Advantages and Disadvantages.

Performance of Lines: Representation of lines, short transmission lines, medium length lines, nominal T and л-representations, long transmission lines. The equivalent circuit representation of a long Line, A, B, C, D constants, Ferranti Effect.

UNIT-III (Transmission & Distribution of Electric Power- II):

Mechanical Design of Over Headlines: Catenary curve, calculation of sag & tension, effects of wind and ice

loading, sag template, vibration dampers

Overhead line Insulators: Type of insulators and their applications, potential distribution over a string of insulators, methods of equalizing the potential, string efficiency

UNIT-IV (Transmission Line Parameters):

Inductance and Capacitance Calculations of Transmission Lines: Line conductors, inductance and capacitance of single-phase and three-phase lines with symmetrical and unsymmetrical spacing, Composite conductors-transposition, bundled conductors, and effect of earth on capacitance.

UNIT-V (Insulated Cables):

Insulated Cables: Introduction, insulation, insulating materials, Extra high voltage cables, grading of cables, insulation resistance of a cable, Capacitance of a single core and three core cables, Overhead lines versus underground cables, types of cables

Textbooks:

1. Kothari & Nagrath, “Power System Engineering”, McGraw-Hill Education
2. B.R. Gupta, “Generation of Electrical Energy”, S. Chand Publication.
3. Chakrabarti A., Soni M.L., Gupta P.V., and Bhatnagar U.S., 'A textbook on Power Systems Engg.', Dhanpat Rai and Sons, New Delhi.
4. JB Gupta, ‘A Course in Power Systems’, S.K. Kataria and Sons.
5. C.L. Wadhwa, “Electrical Power System”, New Age International Ltd. Third Edition.
6. Arun Ingole, "Power Transmission and Distribution", Pearson Education, 2018
7. A. J. Wood & B.F. Wollenburg, “Power Generation, Operation and Control “John Wiley & Sons.

Reference Books:

1. Wadhwa, C.L., ‘Generation Distribution and Utilization of Electrical Energy', New Age International publishers.
2. Deshpande M.V, `Elements of Electrical Power systems Design’, Pitman, New Delhi, PHI Learning Private Limited,
3. S.N. Singh, “Electric Power Generation, Transmission & Distribution”, PHI Learning

BEE502: CONTROL SYSTEM

Course Outcome		Knowledge Level, KL
Upon the completion of the course, the student will be able to:		
CO 1	Obtain transfer functions to predict the correct operation of open loop and closed loop control systems and identify the basic elements, structures and thecharacteristics of feedback control systems.	**K3**
CO 2	Measure and evaluate the performance of basic control systems in time domain. Design specification for different control action.	**K4**
CO 3	Analyze the stability of linear time-invariant systems in time domain using Routh-Hurwitz criterion and root locus technique.	**K4**
CO 4	Determine the stability of linear time-invariant systems in frequency domain using Nyquist criterion and Bode plot.	**K4**
CO 5	Design different type of compensators to achieve the desired performance of control System by root locus and Bode plot method. Develop and analyze the intermediate states of the system using state space analysis.	**K4**

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)
K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create

Unit-I:

Control System Concepts: Elements of control systems, concept of open loop and closed loop systems, Examples and application of open loop and closed loop systems, Mathematical Modelling of Physical Systems (Electro Mechanical), Determination of transfer function by block diagram reduction techniques and signal flow method using Mason's gain formula, Basic Characteristics of negative feedback control systems.

Control System Components: Constructional and working concept of AC & DC servomotor, synchro's, stepper motor and tachometer.

Unit-II:

Time Response Analysis: Standard test signals, time response analysis of first and second order systems time response specifications of second order system for unit step input, location of roots of characteristics equation and corresponding time response, steady state errors and error constants.

Basic modes of feedback control: Proportional, Derivative, Integral and PID controllers, controller design using Ziegler Nichols.

Unit-III:

Stability and Algebraic Criteria: Concept of stability and its necessary conditions, Routh-Hurwitz criteria and its limitations.

Root Locus Technique: Salient features of root locus plot, Procedure for plotting root locus, root contours.

Unit-IV:

Frequency Response Analysis: Frequency Response analysis from transfer function model, Construction of polar and inverse polar plots.

Stability in Frequency Domain: Nyquist stability criterion, Determination of gain and phase margin from Bode & Nyquist Plots, Correlation between time and Frequency Responses.

Unit-V

Introduction to Design: The design problems and preliminary considerations of lead, lag and lead-lag compensation networks, design of closed loop systems using compensation techniques in time and frequency domains.

State Space Technique: Concept of single input and multiple input and multiple output system, State-space model of physical system, conversion of state-space to transfer function model and vice-versa, State transition matrix, Concept of controllability and observability and their testing.

Text Book:

1. I. J. Nagrath & M. Gopal, "Control System Engineering", 6th Ed. New Age International Publishers, 4th Edition 2018
2. M. Gopal, "Control System Principles and Design", McGraw Hill 4th Edition
3. Ogata, "Modern Control Engineering, 5th Edition", Pearson Education, 2015
4. B.C. Kuo & Farid Golnaraghi, "Automatic Control Systems", 10th Edition, McGraw Hill
5. D. Roy Choudhary, "Modern Control Engineering", Prentice Hall of India.
6. A.K. Jairath, "Problems and solutions of Control System with Essential Theory", CBS Publishers

Reference Books:

1. (Schaums Outlines Series) Joseph J. Distefano III, Allen R. Stubberud, Ivan J. Williams, "Control Systems", 3rd Edition, McGraw Hill, Special Indian Edition, 2010.
2. Norman S. Mise, Control System Engineering, Wiley Publishing Co.
3. Ajit K Mandal, "Introduction to Control Engineering" New Age International.
4. R.T. Stefani, B.Shahian, C.J.Savant and G.H. Hostetter, "Design of Feedback Control Systems" Oxford University Press.
5. Samarjit Ghosh, "Control Systems theory and Applications", Pearson Education.

BEE503: ELECTRICAL MACHINE-II

Pre-requisites of course: Basic Electrical Engineering, Electrical Machine-I

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of the course, the student will be able to:		
CO 1	Demonstrate the constructional details and principle of operation of three-phase Induction and Synchronous Machines.	**K3**
CO 2	Analyze the performance of the three-phase Induction and Synchronous Machines using the phasor diagrams and equivalent circuits.	**K4**
CO 3	Select appropriate three-phase AC machine for any application and appraise its significance.	**K4**
CO 4	Start and observe the various characteristics of three phase Induction & Synchronous Machines	**K4**
CO 5	Explain the principle of operation and performance of Single-Phase Induction Motor & Universal Motor.	**K3**

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)

K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6-Create

Detailed Syllabus:

Unit- I: Synchronous Machine-I

Constructional features, Armature winding, EMF Equation, Winding coefficients, Equivalent circuit and phasor diagram, Armature reaction, O. C. & S. C. tests, Voltage regulation using Synchronous Impedance method, MMF method, Potier's Triangle method, Voltage and frequency control (Governer system) of alternators , Parallel operation of synchronous generators, Operation on infinite bus, Synchronizing power and torque co-efficient.

UNIT – II: Synchronous Machine II

Two reaction theory, Transient and sub-transient reactance, Power flow equations of cylindrical and salient pole machines, Operating characteristics. Synchronous Motor - Starting methods, Effect of varying field current at different loads, V- curves, Hunting & damping, Synchronous condenser.

UNIT – III: Three phase Induction Machine - I

Constructional features, Rotating magnetic field, Principle of operation, Phasor diagram, Equivalent circuit, Torque and power equations, Torque- slip characteristics, No load & blocked rotor tests, Efficiency, principle of operation and application of induction generator

UNIT – IV: Three phase Induction Machine- II

Starting, Deep bar and double cage rotors, Cogging & Crawling, Speed control (with and without emf injection in rotor circuit).

UNIT – V: Single phase Induction Motor

Double revolving field theory, Equivalent circuit, No load and blocked rotor tests, Starting methods, Repulsion motor, Universal motor.

Text Books:

1. P S Bimbhra, "Electrical Machinery", Khanna Publisher
2. I J Nagrath & D.P. Kothari, "Electrical Machines", McGraw Hill
3. Irving Kosow, "Electric Machinery and Transformers", 2nd Edition
4. Rajendra Prasad , "Electrical Machines", PHI Learning Pvt. Ltd,
5. AE Fitzgerald, C. Kingsley Jr and Umans, "Electric Machinery", McGraw Hill, International Student Edition.

Reference Books:

1. H. Cotton, "Electrical Technology", CBS Publication.
2. MG Say, "The Performance and Design of AC machines", Pitman& Sons.
3. PS Bimbhra, "Generalized Machine Theory". 4. Samarjit Ghosh, " Electrical Machines", Pearson Education

BEE551: POWER SYSTEM LABORATORY - I

Pre-requisites of course: Basic understanding of Scilab/MATLAB/C/C++

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO1	Use programming tools /Software: Scilab, MATLAB or any C, C++ - Compiler and formulate a program/simulation model for calculation of various parameters related to transmission line.	**K6**

Note: Minimum ten experiments are to be performed from the following list, on a softwareplatform preferably on Scilab, MATLAB, or any C, C++ - Compiler

1. Calculate the parameters of single-phase transmission line
2. Calculate the parameters of three phase single circuit transmission line
3. Calculate the parameters of three phase double circuit transmission line
4. Determine the ABCD constant for transmission line.
5. Simulate the Ferranti effect in transmission line
6. Calculate the corona loss of transmission line
7. Calculation of sag & tension of transmission line
8. Calculation of string efficiency of insulator of transmission line
9. Calculation for grading of underground cables
10. Simulate the skin effect in the transmission line
11. Calculation of ground clearance of transmission line
12. Calculate the parameters for underground cable.
13. Simulation of small PV generation and finding their characteristics.
14. Simulation of small wind generation and finding their characteristics.

Spoken Tutorial (MOOCs):

Spoken Tutorial MOOCs, ' Course on Scilab', IIT Bombay (http://spoken-tutorial.org/)

BEE552: CONTROL SYSTEM LABORATORY

Pre-requisites of course: Basic understanding of Scilab/MATLAB or any equivalent open source software

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Determine the characteristics of control system components like ac servo motor, synchro, potentiometer, servo voltage stabilizer and use them in error detector mode.	**K4**
CO2	Compare the performance of control systems by applying different controllers / compensators.	**K5**
CO3	Analyze the behavior of dc motor in open loop and closed loop conditions at various loads & determine the response of 1st& 2nd order systems for various values of constant K.	**K5**
CO4	Apply different stability methods of time & frequency domain in control systems using software & examine their stability.	**K4**
CO5	Convert the transfer function into state space & vice versa & obtain the time domain response of a second order system for step input and their performance parameters using software.	**K5**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1 – Remember K2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6 – Create

Note: Minimum 10 experiments are to be performed from the following list:

1. To determine speed-torque characteristics of an AC servomotor.
2. To study
   i) Synchro Transmitter characteristics.
   ii) Obtain Synchro Transmitter – Receiver output vs input characteristics.
3. To determine response of first order and second order systems for step input for various values of constant 'K' using linear simulator unit and compare theoretical and practical results.
4. To study characteristics of positional error detector by angular displacement of two servo potentiometers.
5. To simulate and compare the response of 2nd order system with and without lead, lag, Lead- Lag compensator / simulate PID controller for transportation lag.
6. To study P, PI and PID temperature controller for an oven and compare their characteristics.
7. To study performance of servo voltage stabilizer at various loads using load bank.
8. To study behavior of separately excited dc motor in open loop and closed loop conditions at various loads.
9. To determine time domain response of a second order system for step input and obtain performance parameters.
10. To convert transfer function of a system into state space form and vice-versa.
11. To plot root locus diagram of an open loop transfer function and determine range of gain 'k' for stability.
12. To plot a Bode diagram of an open loop transfer function.
13. To draw a Nyquist plot of an open loop transfers functions and examine the stability of the closed loop system.

Software based experiments (Scilab/MATLAB or any equivalent open source software**)**

Spoken Tutorial (MOOCs):

Spoken Tutorial MOOCs, ' Course on Scilab', IIT Bombay (http://spoken-tutorial.org/)

Reference Books:

1. K.Ogata,"Modern Control Engineering" Prentice Hall of India.
2. Norman S.Nise, "Control System Engineering", John Wiley & Sons.
3. M.Gopal, "Control Systems: Principles & Design" Tata McGraw Hill.

BEE553: ELECTRICAL MACHINE-II LABORATORY

Pre-requisites of course: Basic Electrical Engineering Lab**,** Electrical Machine-I Lab.

	Course Outcomes:	**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Perform various tests and demonstrate the various characteristics of three phase induction motor.	**K4**
CO2	Demonstrate the working of three phase synchronous machine under different operating conditions.	**K4**
CO3	Evaluate the performance of single-phase induction motor under different operating conditions.	**K5**
CO4	Develop simulation models for Electrical Machines.	**K6**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1 – Remember K2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6 – Create

Note: Minimum 10 experiments are to be performed from the following list:

1. To perform no load and blocked rotor tests on a three-phase squirrel cage induction motor and determine equivalent circuit.
2. To perform a load test on a three-phase induction motor and draw Torque -speedcharacteristics
3. To perform no load and blocked rotor tests on a single phase induction motor and determine equivalent circuit.
4. To study speed control of three phase induction motor by varying supply voltage and by keeping V/f ratio constant.
5. To perform open circuit and short circuit tests on a three phase alternator.
6. To determine V-curves and inverted V-curves of a three phase synchronous motor.
7. To determine the direct axis reactance (Xd) and quadrature axis reactance (Xq) of synchronous machine.
8. To study synchronization of an alternator with the infinite bus by using: (i) dark lamp method (ii) two bright and one dark lamp method.
9. To determine speed-torque characteristics of three phase slip ring induction motor and study the effect of including resistance, or capacitance in the rotor circuit.
10. To determine speed-torque characteristics of single phase induction motor and study the effect of voltage variation.
11. To determine speed-torque characteristics of a three phase induction motor by (i) keeping v/f ratio constant (ii) increasing frequency at the rated voltage.
12. To draw O.C. and S.C. characteristics of a three phase alternator from the experimental data and determine voltage regulation at full load, and unity, 0.8 lagging and leading power factors.
13. To determine steadystate performance of a three phase induction motor using equivalent circuit.
14. Load Test on Three Phase Alternator.

***The available experiments from above list may be performed on virtual lab on following virtual lab link:**http://vlab.co.in/

BEE052: SENSORS AND TRANSDUCERS

Pre-requisites of course: Basic Electrical Engineering, Basic signals & systems

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Understand the working of commonly used sensors in industry for measurement of displacement, force and pressure.	**K3**
CO2	Recognize the working of commonly used sensors in industry for measurement of temperature, position, accelerometer, vibration sensor, flow and level.	**K3**
CO3	Identify the application of machine vision.	**K2**
CO4	Conceptualize signal conditioning and data acquisition methods.	**K2**
CO5	Comprehend smart sensors and their applications in automation systems.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1 – Remember K_2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6 – Create

Detailed Syllabus:

Unit- I:

Sensors & Transducer: Definition, Classification of transducers, Advantages and Disadvantages of Electrical Transducers; Measurement of displacement using Potentiometer, LVDT & Optical Encoder; Measurement of force using strain gauges & load cells; Measurement of pressure using LVDT based diaphragm & piezoelectric sensor.

Unit-II:

Measurement of temperature using Thermistors, Thermocouples & RTD, Concept of thermal imaging; Measurement of position using Hall effect sensors; Proximity sensor: Inductive, Capacitive & Photoelectric, Use of proximity sensor as accelerometer and vibration sensor; Flow Sensor: Ultrasonic & Laser; Level Sensor: Ultrasonic & Capacitive.

Unit -III:

Machine Vision: Introduction to machine vision, Difference between machine vision and computer vision; Imaging Sensors: CCD and CMOS; sensing & digitizing function in machine vision, image processing and analysis, training the vision system in a pick and place robot.

Unit-IV:

Signal Conditioning: Introduction, Functions of signal conditioning equipment, need for amplification of signals, Types of amplifiers.

Data Acquisition Systems and Conversion: Introduction, Objectives & configuration of data acquisition system, Analog & Digital IO, Counters, Timers, need of data conversion.

Unit V:

Smart Sensors: General Structure of smart sensors & its components, Characteristic of smart sensors: Self calibration, Self-testing & self-communicating, Application of smart sensors: Smart city, Industrial robots & electric vehicles, body temperature and blood oxygen measurement using smart sensors.

Text Books:

1. A K Sawhney, "Electrical & Electronic Measurement & Instrument", Dhanpat Rai & Sons, India.
2. DVS Murthy, Transducers and Instrumentation, PHI 2nd Edition 2013
3. S. Gupta, J.P. Gupta / PC interfacing for Data Acquisition & Process Control, 2nd ED / Instrument Society of America, 1994.

4. D Patranabis, Sensors and Transducers, PHI 2nd Edition 2013.

Reference Books:

1. Arun K. Ghosh, Introduction to measurements and Instrumentation, PHI, 4th Edition 2012.
2. A.D. Helfrick and W.D. cooper, Modern Electronic Instrumentation & Measurement Techniques, PHI – 2001
3. Hermann K.P. Neubert, "Instrument Transducers" 2nd Edition 2012, Oxford University Press.

BEE053: INDUSTRIAL AUTOMATION & CONTROL

Pre-requisites of course: Digital Electronics

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO1	Understand the concept of automation, its terminology and basic communication protocol.	**K2**
CO2	Apply Relay logic for automation.	**K3**
CO3	Learn about PLC, its operation and application in automation.	**K3**
CO4	Analyze the industrial sensors, its terminology and how one can interface with PLC.	**K3**
CO5	Demonstrate Pneumatic system and its application in industry.	**K3**

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)
K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create

Detailed Syllabus:

Unit1: Introduction of Automation system
Introduction to Industrial Automation, Requirement of automation systems, Application areas, Architecture of Industrial Automation system, Introduction of PLC and supervisory control and data acquisition (SCADA). Industrial communication protocols: modbus & Profibus

Unit2: Automation using relay logic
Relay Circuits: Construction & Principle of Operation, Types of Relays, Relay as a memory element, Contactor Circuits, Advantages of Contactors over Relay, DOL circuit implementation using contactor, Automation problems based on relays, PLC Introduction: History & Current Trends, Basic Block Diagram of PLC, Classification of PLCs

Unit3: Automation using PLC
Types of PLC I/O: Analog and Digital, Sink and Source concept, PLC programming: Ladder diagram, Sequential flow chart, ladder programming, Timer instructions – On delay, Off delay, Cyclic and Retentive timers, Up /Down Counters, math instructions

Unit4: Industrial sensors and its application
Introduction to Industrial sensors: optical, inductive, capacitive Sensors, PNP and NPN senor concept, interfacing of sensors with PLC, 4-20 ma current loops, HART protocol, modes of HART protocol

Unit5: Basics of Pneumatics and its use in automation
Introduction to Pneumatics, Role in industries, Laws : Boyel's law, Charle's Law Bernoulli Equation, Humidity (Absolute & Relative) , Dew Point (ADP, PDP) Basic, Pneumatic System (Compressor, After coolers, Dryers, Air Tank, Service Unit (FRL), Actuators(single acting, double acting), Valves : 2/2 & 3/2 Valves ,Problems based on valves and actuator

Text Books:

1. Industrial Instrumentation and Control, by Singh, McGraw Hill.
2. Programmable Logic Controllers with Control Logix, by Jon Stenerson, Delmar Publishers, 2009
3. Webb John W. and Reis A. Ronald, "Programmable Logic Controllers Principles and Applications" PHI ,New Delhi, Latest edition
4. Bolton W, "Programmable Logic Controllers" Elsevier India Pvt. Ltd. New Delhi

Reference Books:

1. B. Pneumatic Systems-Principles and Maintenance Mazumdar S. R
2. John R Hackworth, "Programmable Logic Controllers" Pearson Education New Delhi, Latest edition

BEE055: OPTIMIZATION TECHNIQUES

Pre-requisites of course: Basic mathematics

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Understand the importance of optimization techniques in engineering applications	**K2**
CO2	Learn optimization methods for solving linear programming problems	**K3**
CO3	Learn optimization methods for solving nonlinear programming problems	**K3**
CO4	Be aware of the concept of simulation and modern methods of optimization	**K3**
CO5	Apply optimization techniques to electrical engineering problems	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1 – Remember K_2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6 – Create

Detailed syllabus:

Unit I

Introduction to Optimization: Engineering application of Optimization, Statement of an optimization problem, Optimal problem formulation, Classification of optimization problem, Optimum design concepts: Definition of Global and Local optima using basic calculus concepts; Classical Optimization Techniques: Unconstrained Optimization - Single variable optimization, Constrained multivariable optimization with equality constraints - Lagrange multipliers method, Constrained multivariable optimization with inequality constraints - Kuhn-Tucker conditions.

Unit II

Linear Programming: Standard form of linear programming, Graphical solution, Simplex method, Big-M method, Duality theory, Decomposition principle, Transportation problem using North-West Corner rule and Least cost rule.

Unit III

Non-Linear Programming: Standard form of non-linear programming, One–Dimensional Minimization Methods - Unimodal function, Dichotomous search, interval halving method; Unconstrained Optimization Techniques - Univariate method, Steepest descent method; Constrained Optimization Techniques - Interior Penalty function method, Exterior penalty function method.

Unit IV

Simulation: Definition, types of simulation, General process of simulation, advantages & disadvantages of simulation.

Project Management Techniques: PERT and CPM

Modern methods of Optimization: Genetic algorithm, working principle, fitness function, GA operators – crossover & mutation, comparison of GA with traditional methods.

Unit V

Case study (algorithm only): Economic load scheduling of power plant (without considering losses), maintenance scheduling of machines in manufacturing industry, fuzzy logic based speed control of DC machines.

Text Books:

1. S.S.Rao,"Optimization - Theory and Applications", Wiley-Eastern Limited.

2. Ronald L. Rardin, "Optimization in Operations Research", Pearson Education, 2019
3. D.E. Goldberg, "Genetic Algorithm in Search Optimization and Machine Learning ", Addision-Wesley Publication, 1989
4. Kwang Y. Lee, Mohamed A. El-Sharkawi, “Modern heuristic optimization techniques, Theory and applications to power systems”, Wiley-Interscience

Reference Books:

1. David G. Luenberger, "Introduction of Linear and Non-Linear Programming ", Wesley Publishing Company.
2. Polak,"Computational methods in Optimization ", Academic Press.
3. Pierre D.A., "Optimization Theory with Applications", Wiley Publications.
4. Kalyanmoy deb,"Optimization for Engineering Design: Algorithms and Examples", PHI Publication
5. L.P. Singh, "Advanced Power System Analysis and Dynamics ", Wiley Eastern Limited.

BEEZ051: Operating System		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of the course, the student will be able to:		
CO 1	Understand the structure and functions of OS	K1, K2
CO 2	Learn about Processes, Threads and Scheduling algorithms.	K1, K2
CO 3	Understand the principles of concurrency and Deadlocks	K2
CO 4	Learn various memory management schemes.	K2
CO 5	Study I/O management and File systems.	K2, K4
DETAILED SYLLABUS		**3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction**: Operating system and functions, Classification of Operating systems- Batch, Interactive, Time sharing, Real-Time System, Multiprocessor Systems, Multiuser Systems, Multiprocess Systems, Multithreaded Systems, Operating System Structure- Layered structure, System Components, Operating System services, Reentrant Kernels, Monolithic and Microkernel Systems.	**08**
II	**Concurrent Processes:** Process Concept, Principle of Concurrency, Producer / Consumer Problem, Mutual Exclusion, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Test and Set operation; Classical Problem in Concurrency- Dining Philosopher Problem, Sleeping Barber Problem; Inter Process Communication models and Schemes, Process generation.	**08**
III	**CPU Scheduling:** Scheduling Concepts, Performance Criteria, Process States, Process Transition Diagram, Schedulers, Process Control Block (PCB), Process address space, Process identification information, Threads and their management, Scheduling Algorithms, Multiprocessor Scheduling. Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock.	**08**
IV	**Memory Management:** Basic bare machine, Resident monitor, Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Cache memory organization, Locality of reference.	**08**
V	**I/O Management and Disk Scheduling:** I/O devices, and I/O subsystems, I/O buffering, Disk storage and disk scheduling, RAID. File System: File concept, File organization and access mechanism, File directories, and File sharing, File system implementation issues, File system protection and security.	**08**

Text Books:

1. Silberschatz, Galvin and Gagne, "Operating Systems Concepts", Wiley
2. Sibsankar Halder and Alex A Aravind, "Operating Systems", Pearson Education
3. Harvey M Dietel, " An Introduction to Operating System", Pearson Education
4. D M Dhamdhere, "Operating Systems : A Concept based Approach", 2nd Edition, TMH
5. William Stallings, "Operating Systems: Internals and Design Principles", 6th Edition, Pearson Education

BCS051: Statistical Computing		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to:		
CO 1	Understand and apply the probability distributions, random number generation and density estimations to perform analysis of various kinds of data.	K2, K_4, K_6
CO 2	Understand and manipulate data, design and perform simple Monte Carlo experiments, and be able to use resampling methods.	K_5, K_6
CO 3	Perform statistical analysis on variety of data.	K_2, K_5
CO 4	Perform appropriate statistical tests using R and visualize the outcome.	K_2, K_4
CO 5	Discuss the results obtained from their analyses after creating customized graphical and numerical summaries.	K_2, K_3
DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Descriptive Statistics:** Diagrammatic representation of data, measures of central tendency, measures of dispersion, measures of skewness and kurtosis, correlation, inference procedure for correlation coefficient, bivariate correlation, multiple correlations, linear regression and its inference procedure, multiple regression. **Probability:** Measures of probability, conditional probability, independent event, Bayes' theorem, random variable, discrete and continuous probability distributions, expectation and variance, markov inequality, chebyshev's inequality, central limit theorem.	**08**
II	**Inferential Statistics:** Sampling & Confidence Interval, Inference & Significance. Estimation and Hypothesis Testing, Goodness of fit, Test of Independence, Permutations and Randomization Test, t- test/z-test (one sample, independent, paired), ANOVA, chi-square. **Linear Methods for Regression Analysis:** multiple regression analysis, orthogonalization by Householder transformations (QR); singular value decomposition (SVD); linear dimension reduction using principal component analysis (PCA).	**08**
III	**Pseudo-Random Numbers:** Random number generation, Inverse-transform, acceptance-rejection, transformations, multivariate probability calculations. **Monte Carlo Integration:** Simulation and Monte Carlo integration, variance reduction, Monte Carlo hypothesis testing, antithetic variables/control variates, importance sampling, stratified sampling Markov chain Monte Carlo (McMC): Markov chains; Metropolis-Hastings algorithm, Gibbs sampling; convergence	**08**
IV	**Resampling Methods:** Cross-validation, Bootstrapping, Jackknife resampling, percentile confidence intervals, permutation tests **Density Estimation:** Univariate density estimation, kernel smoothing, multivariate density estimation **Numerical Methods:** Root finding; more on numerical integration; numerical maximization/minimization; constrained and unconstrained optimization; EM (Expectation-	**08**

	Maximization) algorithm; simplex algorithm	
V	**Introduction to R programming:** History of R programming, starting and ending R, R as a scientific calculator , handling package, workspace, inspecting variables, operators and expressions inR, data objects and types, vectors, matrices and arrays, lists and data frames, built-in and user-defined functions , strings and factors, flow control and loops, advanced looping, date and times. **Using R for statistical analysis:** Importing data files, exporting data, outputting results, exporting graphs, graphics in R, interactively adding information of plot, performing data analysis tasks. R commands for descriptive statistics, data aggregation, representation of multivariate data, code factorization and optimization, statistical libraries in R.	**08**

References:

1. S.C. Gupta & V.K. Kapoor, "Fundamentals of Mathematical Statistics", Sultan Chand & Sons
2. Sheldon M. Ross, "Introduction to Probability and Statistics for Engineers and Scientists", Academic Press.
3. Dudewicz, E.J., Mishra, S.N., "Modern Mathematical Statistics", Willy
4. Purohit S. G., Gore S. D., Deshmukh S. K., "Statistics using R, Narosa
5. Rizzo, M. L., "Statistical Computing with R", Boca Raton, FL: Chapman & Hall/CRC Press
6. Normal Maltoff, The Art of R programming, William
7. Dalgaard, Peter, "Introductory statistics with R", Springer Science & Business Media
8. M. D. Ugarte, A. F. Militino, A. T. Arnholt, "Probability and Statistics with R", CRC Press
9. Kundu, D. and Basu, A., "Statistical computing – existing methods and recent developments", Narosa
10. Gentle, James E., Härdle, Wolfgang Karl, Mori, Yuich, "Handbook of Computational Statistics", Springer
11. Givens and Hoeting, "Computational Statistics", Wiley Series in Prob. and Statistics
12. Michael J. Crawley "The R Book", John Wiley and Sons.
13. Richard Cotton, "Learning R", O'Reilly
14. Brain S. Everitt, "A Handbook of Statistical Analysis Using R", Second Edition, LLC
15. Randall E. Schumacker, "Learning Statistics Using R", Sage.
16. Jared P. Lander, "R for Everyone" Addison Wesley.
17. Monahan, J.F., "Numerical methods of statistics", Cambridge University Press.
18. Robert, C. and Casella, G., "Introducing Monte Carlo Methods with R", Springer Verlag, New York.

BCS053: Computer Graphics		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to:		
CO 1	Understand the graphics hardware used in field of computer graphics.	K_2
CO 2	Understand the concept of graphics primitives such as lines and circle based on different algorithms.	K_2, K_4
CO 3	Apply the 2D graphics transformations, composite transformation and Clipping concepts.	K_4
CO 4	Apply the concepts of and techniques used in 3D computer graphics, including viewing transformations.	K_2, K_3
CO 5	Perform the concept of projections, curve and hidden surfaces in real life.	K_2, K_3
DETAILED SYLLABUS		**3-0-0**
Unit	**Topics**	**Proposed Lecture**
I	**Introduction and Line Generation:** Types of computer graphics, Graphic Displays-Random scan displays, Raster scan displays, Frame buffer and video controller, Points and lines, Line drawing algorithms, Circle generating algorithms, Mid-point circle generating algorithm, and parallel version of these algorithms.	**08**
II	**Transformations:** Basic transformation, Matrix representations and homogenous coordinates, Composite transformations, Reflections and shearing. **Windowing and Clipping:** Viewing pipeline, viewing transformations, 2-D Clipping algorithms- Line clipping algorithms such as Cohen Sutherland line clipping algorithm, Liang Barsky algorithm, Line clipping against non-rectangular clip windows; Polygon clipping – Sutherland Hodgeman polygon clipping, Weiler and Atherton polygon clipping, Curve clipping, Text clipping	**08**
III	**Three Dimensional:** 3-D Geometric Primitives, 3-D Object representation, 3-D Transformation, 3- D viewing, projections, 3-D Clipping.	**08**
IV	**Curves and Surfaces:** Quadric surfaces, Spheres, Ellipsoid, Blobby objects, Introductory concepts of Spline, Bspline and Bezier curves and surfaces.	**08**
V	**Hidden Lines and Surfaces:** Back Face Detection algorithm, Depth buffer method, A-buffer method, Scan line method, basic illumination models– Ambient light, Diffuse reflection, Specular reflection and Phong model, Combined approach, Warn model, Intensity Attenuation, Color consideration, Transparency and Shadows.	**08**

Text books:

1. Donald Hearn and M Pauline Baker, “Computer Graphics C Version”, Pearson Education
2. Foley, Vandam, Feiner, Hughes – “Computer Graphics principle”, Pearson Education.
3. Rogers, “ Procedural Elements of Computer Graphics”, McGraw Hill
4. W. M. Newman, R. F. Sproull – “Principles of Interactive computer Graphics” – McGraw Hill.
5. Amrendra N Sinha and Arun D Udai,” Computer Graphics”, McGraw Hill.
6. R.K. Maurya, “Computer Graphics ” Wiley Dreamtech Publication.
7. Mukherjee, Fundamentals of Computer graphics & Multimedia, PHI Learning Private Limited.
8. Donald Hearn and M Pauline Baker, “Computer Graphics with Open GL”, Pearson education

BEEZ052: Object Oriented Programming with Java		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to:		
CO 1	Develop the object-oriented programming concepts using Java	K_3, K_4
CO 2	Implement exception handling, file handling, and multi-threading in Java	K_2,K_4
CO 3	Apply new java features to build java programs.	K_3
CO 4	Analyse java programs with Collection Framework	K_4
CO 5	Test web and RESTful Web Services with Spring Boot using Spring Framework concepts	K_5
DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction:** Why Java, History of Java, JVM, JRE, Java Environment, Java Source File Structure, and Compilation. Fundamental, **Programming Structures in Java:** Defining Classes in Java, Constructors, Methods, Access Specifies, Static Members, Final Members, Comments, Data types, Variables, Operators, Control Flow, Arrays & String. **Object Oriented Programming**: Class, Object, Inheritance Super Class, Sub Class, Overriding, Overloading, Encapsulation, Polymorphism, Abstraction, Interfaces, and Abstract Class. **Packages:** Defining Package, CLASSPATH Setting for Packages, Making JAR Files for Library Packages, Import and Static Import Naming Convention For Packages	**08**
II	**Exception Handling**: The Idea behind Exception, Exceptions & Errors, Types of Exception, Control Flow in Exceptions, JVM Reaction to Exceptions, Use of try, catch, finally, throw, throws in Exception Handling, In-built and User Defined Exceptions, Checked and Un-Checked Exceptions. **Input /Output Basics**: Byte Streams and Character Streams, Reading and Writing File in Java. **Multithreading:** Thread, Thread Life Cycle, Creating Threads, Thread Priorities, Synchronizing Threads, Inter-thread Communication.	**08**
III	**Java New Features:** Functional Interfaces, Lambda Expression, Method References, Stream API, Default Methods, Static Method, Base64 Encode and Decode, ForEach Method, Try-with-resources, Type Annotations, Repeating Annotations, Java Module System, Diamond Syntax with Inner Anonymous Class, Local Variable Type Inference, Switch Expressions, Yield Keyword, Text Blocks, Records, Sealed Classes	**08**
IV	**Java Collections Framework:** Collection in Java, Collection Framework in Java, Hierarchy of Collection Framework, Iterator Interface, Collection Interface, List Interface, ArrayList, LinkedList, Vector, Stack, Queue Interface, Set Interface,	**08**

	HashSet, LinkedHashSet, SortedSet Interface, TreeSet, Map Interface, HashMap Class, LinkedHashMap Class, TreeMap Class, Hashtable Class, Sorting, Comparable Interface, Comparator Interface, Properties Class in Java.	
V	**Spring Framework:** Spring Core Basics-Spring Dependency Injection concepts, Spring Inversion of Control, AOP, Bean Scopes- Singleton, Prototype, Request, Session, Application, Web Socket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles **Spring Boot:** Spring Boot Build Systems, Spring Boot Code Structure, Spring Boot Runners, Logger, BUILDING RESTFUL WEB SERVICES, Rest Controller, Request Mapping, Request Body, Path Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications	**08**

Text Books

1. Herbert Schildt, "Java The complete reference", McGraw Hill Education
2. Craig Walls, “Spring Boot in Action” Manning Publication
1. Steven Holzner, “Java Black Book”, Dreamtech.
2. Balagurusamy E, “Programming in Java”, McGraw Hill
3. Java: A Beginner’s Guide by Herbert Schildt, Oracle Press
4. Greg L. Turnquist “Learning Spring Boot 2.0 - Second Edition”, Packt Publication
1. AJ Henley Jr (Author), Dave Wolf, “Introduction to Java Spring Boot: Learning by Coding”, Independently Published

BCS056: Application of Soft Computing		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to:		
CO 1	Recognize the feasibility of applying a soft computing methodology for a particular Problem	K_2, K_4
CO 2	Know the concepts and techniques of soft computing and foster their abilities in designing and implementing soft computing based solutions for real-world and engineering problems.	K_4, K_6
CO 3	Apply neural networks to pattern classification and regression problems and compare solutions by various soft computing approaches for a given problem.	K_3, K_5
CO 4	Apply fuzzy logic and reasoning to handle uncertainty and solve engineering problems	K_3, K_4
CO 5	Apply genetic algorithms to combinatorial optimization problems	K_3
DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Neural Networks-I (Introduction & Architecture) :** Neuron, Nerve structure and synapse, Artificial Neuron and its model, activation functions, Neural network architecture: single layer and multilayer feed forward networks, recurrent networks. Various learning techniques; perception and convergence rule, Auto-associative and hetro-associative memory.	**08**
II	**Neural Networks-II (Back propagation networks):** Architecture: perceptron model, solution, single layer artificial neural network, multilayer perception model; back propagation learning methods, effect of learning rule co-efficient; back propagation algorithm, factors affecting backpropagation training, applications.	**08**
III	**Fuzzy Logic-I (Introduction):** Basic concepts of fuzzy logic, Fuzzy sets and Crisp sets, Fuzzy set theory and operations, Properties of fuzzy sets, Fuzzy and Crisp relations, Fuzzy to Crisp conversion.	**08**
IV	**Fuzzy Logic –II (Fuzzy Membership, Rules)** : Membership functions, interference in fuzzy logic, fuzzy if-then rules, Fuzzy implications and Fuzzy algorithms, Fuzzyfications & Defuzzificataions, Fuzzy Controller, Industrial applications	**08**
V	**Genetic Algorithm(GA):** Basic concepts, working principle, procedures of GA, flow chart of GA, Genetic representations, (encoding) Initialization and selection, Genetic operators, Mutation, Generational Cycle, applications.	**08**

Text books:

1. S. Rajsekaran & G.A. Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications" Prentice Hall of India.
2. N. P. Padhy," Artificial Intelligence and Intelligent Systems" Oxford University Press. Reference Books:
3. Siman Haykin," Neural Networks" 3rd Edition Pearson Education
4. Timothy J. Ross, "Fuzzy Logic with Engineering Applications" Wiley India.
5. Kumar Satish, "Neural Networks" McGraw Hill

Syllabus

B.Tech. 3rd Year

VI Semester

BEE602: MICROPROCESSOR

Pre-requisites of course: Digital Electronics, Computer Basics

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Demonstrate the general architecture of 8085 & explain its operations, the basic programming instructions of 8085 microprocessor	K2
CO2	Demonstrate the internal architecture of 8085, explain the instruction sets of 8085	K2
CO3	Demonstrate the general architecture of 8086, explain the instruction sets of 8086.	K2
CO4	Explain the basic programming instructions of 8085 microprocessor and write assembly level programs.	K3
CO5	Interface different external peripheral devices the 8085 microprocessor.	K2

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)

K_1 – Remember K_2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6 – Create

Detailed Syllabus:

Unit-I: Introduction to Microprocessor:

Introduction to Microprocessor and its applications, Microprocessor Evolution Tree, Microprocessor Architecture (Harward & Princeton), General Architecture of the Microprocessor and its operations, Component of Microprocessor system: Processor, Buses, Memory, Inputs-outputs (I/Os) and other Interfacing devices.

Unit-II: 8-bit Microprocessor:

Intel 8085 microprocessor: Pin Diagram, Internal architecture: ALU, Registers, Timing and control unit, interrupt.

Instruction Set of 8085: Instruction format, op-codes, mnemonics, no. of bytes, computation of the instruction, Machine cycles and T- states and Execution time computation of an instruction. Classification of instruction with their examples. Writing of assembly Language programs.

Unit-III: 16-bit Microprocessor:

Architecture of Intel 8086: Pin Diagram, Bus Interface Unit, Execution unit, Register organization, Memory addressing, Memory Segmentation, Pipelining, Min & Max operating Modes.

8086 Instruction set: Format, Addressing Modes, Instruction Set Groups: Data transfer, Arithmetic, Logic, String, Branch control transfer and Processor control.

Interrupts: Hardware and software interrupts.

UNIT-IV: Fundamental of Programming:

Program structure for microprocessors, Flowcharts of series, parallel, and control structures.

Assembler Level Programming: Memory space allocation for monitor and user program. Assembly language program using Debug or MASM assembler.

UNIT-V: Peripheral Interfacing:

Programmed I/O, Memory Mapped I/O, Interrupt Driven I/O, DMA I/O interface, Serial and Parallel communications.

Peripheral Devices:

DMA controller (Intel 8237), Programmable peripheral interface (Intel 8255), Programmable timer/counter (Intel 8253/8254), Programmable Interrupt Controller (Intel 8259).

Text Books:

1. Gaonkar, Ramesh S, "Microprocessor Architecture, programming and applications with the 8085" Pen ram International Publishing 5th Ed.
2. Avtar Singh& Walter A. Triebel "8088 & 8086 Microprocessor" Pearson Education.
3. Ray, A.K. & Burchandi, K.M., "Advanced Microprocessors and Peripherals: Architecture, Programaming and Interfacing" Tata Mc. Graw Hill.
4. AK Gautam, "Advanced Microprocessors", Khanna Publishers.

Reference Books:

1. Brey, Barry B. "INTEL Microprocessors" Prentice Hall (India)
2. Aditya P Mathur, "Introduction to Microprocessor" Tata McGraw Hill
3. M. Rafiquzzaman, "Microprocessors- Theory & applications", Pearson India.
4. B. Ram, "Advanced Microprocessor & Interfacing" Tata McGraw Hill
5. Renu Singh & B.P. Singh, "Microprocessor and Interfacing and applications" New Age International
6. Liu and Gibson G.A., "Microcomputer Systems: The 8086/8088 Family Architecture Programming & Design" Pearson India.

BEE603: POWER ELECTRONICS

Pre-requisites of course: Basic Electrical Engineering, Network Analysis & Synthesis

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Demonstrate the characteristics as well as the operation of BJT, MOSFET, IGBT, SCR, TRIAC and GTO and identify their use in the power switching applications.	**K4**
CO2	Comprehend the non-isolated DC-DC converters and apply their use in different Power electronics applications.	**K3**
CO3	Analyze the phase controlled rectifiers and evaluate their performance parameters.	**K5**
CO4	Apprehend the working of single-phase ac voltage controllers, cyclo-converters and their various applications.	**K3**
CO5	Explain the single-phase and three phase bridge inverters differentiate between CSI and VSI and apply PWM for harmonic reduction.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1 – Remember K_2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6 – Create

Detailed Syllabus:

Unit-I: Power semiconductor devices:
Introduction: Concept of Power Electronics, scope and applications, desired Characteristics of controllable switches
Power semiconductor switches and their characteristics: Power Diode, Power BJT, Power MOSFET, IGBT, SCR, TRIAC, GTO.

Unit-II:
Thyristor: Rating & protection, Methods of SCR commutation, Gate Drive Circuit, Series and Parallel operation.
DC-DC Converters: Introduction, Control Strategies, Buck converter, Boost Converter, Buck-Boost converter, Analysis of buck converter, introduction to Switched Mode power Supply (SMPS).

Unit-III: Phase Controlled Converters:
Single phase half wave controlled rectifier with various loads, Effect of freewheeling diode, Single phase fully controlled and half controlled bridge converters with various loads. Performance Parameters of single phase uncontrolled and controlled converters, three phase half wave converters, Three phase fully controlled and half controlled bridge converters, Effect of source impedance on Single phase converters.

Unit-IV: AC Voltage Controllers:
Principle of On-Off and phase controls, Single phase ac voltage controller with resistive and inductive loads, sequence control, Introduction to Matrix converter.
Cyclo Converters: Basic principle of operation, single phase to single phase, three phase to single phase output voltage equation.

Unit-V: Inverters:
Single phase and Three phase bridge inverters, voltage source inverters, current source inverters, Voltage control of single phase inverters, Pulse width modulation, Introduction to Multi level inverter.

Text Books:

1. M.H. Rashid,"Power Electronics: Circuits, Devices & Applications", Pearson Education.
2. Ned Mohan, T.M.Undeland and W.P.Robbins, “Power Electronics: Converters, Applications and Design”, Wiley India Ltd,2008
3. P.C. Sen, “Power Electronics”, McGraw Hill Education (India) Pvt. Ltd.
4. P.S. Bhimbra, “Power Electronics”, Khanna Publishers.

Reference Books:

1. M.S. Jamil Asghar, “Power Electronics” Prentice Hall of India Ltd., 2004
2. Chakrabarti & Rai, “Fundamentals of Power Electronics & Drives”Dhanpat Rai & Sons.
3. V.R. Moorthy, “Power Electronics: Devices, Circuits and Industrial Applications” Oxford University Press,2007
4. S.N.Singh, “A Text Book of Power Electronics” Dhanpat Rai & Sons

BCS603: Computer Networks		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO 1	Explain basic concepts, OSI reference model, services and role of each layer of OSI model and TCP/IP, networks devices and transmission media, Analog and digital data transmission	K_2
CO 2	Apply channel allocation, framing, error and flow control techniques.	K_3
CO 3	Describe the functions of Network Layer i.e. Logical addressing, subnetting & Routing Mechanism.	K_3
CO 4	Explain the different Transport Layer function i.e. Port addressing, Connection Management, Error control and Flow control mechanism.	K_3
CO 5	Explain the functions offered by session and presentation layer and their Implementation.	K_3
CO 6	Explain the different protocols used at application layer i.e. HTTP, SNMP, SMTP, FTP, TELNET and VPN.	K_2
DETAILED SYLLABUS		**3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introductory Concepts**: Goals and applications of networks, Categories of networks, Organization of the Internet, ISP, Network structure and architecture (layering principles, services, protocols and standards), The OSI reference model, TCP/IP protocol suite, Network devices and components. **Physical Layer:** Network topology design, Types of connections, Transmission media, Signal transmission and encoding, Network performance and transmission impairments, Switching techniques and multiplexing.	**08**
II	**Link layer:** Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols). Medium Access Control and Local Area Networks: Channel allocation, Multiple access protocols, LAN standards, Link layer switches & bridges (learning bridge and spanning tree algorithms).	**08**
III	**Network Layer:** Point-to-point networks, Logical addressing, Basic internetworking (IP, CIDR, ARP, RARP, DHCP, ICMP), Routing, forwarding and delivery, Static and dynamic routing, Routing algorithms and protocols, Congestion control algorithms, IPv6.	**08**
IV	**Transport Layer:** Process-to-process delivery, Transport layer protocols (UDP and TCP), Multiplexing, Connection management, Flow control and retransmission, Window management, TCP Congestion control, Quality of service.	**08**

V	**Application Layer:** Domain Name System, World Wide Web and Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Remote login, Network management, Data compression, Cryptography – basic concepts.	**08**

Text books and References:

1. Behrouz Forouzan, “Data Communication and Networking”, McGraw Hill
2. Andrew Tanenbaum “Computer Networks”, Prentice Hall.
3. William Stallings, “Data and Computer Communication”, Pearson.
4. Kurose and Ross, “Computer Networking- A Top-Down Approach”, Pearson.
5. Peterson and Davie, “Computer Networks: A Systems Approach”, Morgan Kaufmann
6. W. A. Shay, “Understanding Communications and Networks”, Cengage Learning.
7. D. Comer, “Computer Networks and Internets”, Pearson.
8. Behrouz Forouzan, “TCP/IP Protocol Suite”, McGraw Hill.

BEE652: MICROPROCESSOR LAB

Pre-requisites of course: Digital Electronics, Computer Basics

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Study of microprocessor system.	K2
CO2	Development of flow chart for understanding the data flow.	K3
CO3	Learning assembly language to program microprocessor-based system.	K3
CO4	Interfacing different peripheral devices with the microprocessor.	K4
CO5	Building logic for microprocessor-based system.	K4

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)
K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create

Note: Minimum ten experiments are to be performed from the following list (on 8085 / 8086 microprocessor) Hardware Based Experiments:

1. To study 8085 / 8086 based microprocessor system.
2. To perform mathematical operations (addition & subtraction) on two 8-bit numbers.
3. To perform multiplication on two 8-bit numbers.
4. To perform division on two 8-bit numbers.
5. To develop and run a program for finding out the largest number from given two 8-bit numbers.
6. To develop and run a program for finding out the smallest number from given two 8-bit numbers.
7. To develop and run a program for arranging in ascending order of a given set of 8-bit numbers.
8. To develop and run a program for arranging in descending order of a given set of 8-bit numbers.
9. To perform conversion of temperature from degree F to degree C.
10. To perform computation of square root of a given number.
11. To obtain interfacing of 8255 – PPI with 8085 microprocessor.
12. To perform microprocessor based traffic light control.
13. To perform microprocessor based stepper motor operation through 8085 / 8086 kit.
14. To obtain interfacing of DMA controller with 8085 / 8086 microprocessor.

PART B SUGGESTIVE LIST OF EXPERIMENTS (Through Virtual Lab Link):

1. Write a program using 8085 Microprocessor for Decimal, Hexadecimal addition and subtraction of two Numbers. (Through Virtual Lab Link).
2. Write a program using 8085 Microprocessor for addition and subtraction of two BCD numbers. (Through Virtual Lab Link)
3. To perform multiplication and division of two 8 bit numbers using 8085. (Through Virtual Lab Link)
4. To find the largest and smallest number in an array of data using 8085 instruction set.

5. To write a program using 8086 to arrange an array of data in ascending and descending order. (Through Virtual Lab Link)
6. To convert given Hexadecimal number into its equivalent ASCII number and vice versa using
   8086 instruction set.
7. To convert given Hexadecimal number into its equivalent BCD number and vice versa using 8086 instruction set.
8. To interface 8253 programmable interval timer and verify the operation of 8253 in six different modes.
9. To write a program to initiate 8251 and to check the transmission and reception of character.
10. Serial communication between two 8085 through RS-232 C port.

Virtual Lab Link: http://vlabs.iitb.ac.in/vlabs-dev/labs_local/microprocessor/labs/explist.php

Available on: http://www.vlab.co.in/broad-area-electronics-and-communications

BEE653: POWER ELECTRONICS LAB

Pre-requisites of course: Basic Electrical Engineering, Network Analysis & Synthesis

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Demonstrate the characteristics and triggering of IGBT, MOSFET, Power transistor and SCR.	**K3**
CO2	Analyze the performance of single phase fully controlled bridge rectifiers under different loading conditions.	**K4**
CO3	Develop simulation models of power electronic circuits.	**K5**

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)
K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create

Note: Minimum 10 experiments are to be performed from the following list:.

1. To study triggering of (i) IGBT (ii) MOSFET (iii) power transistor
2. To study V-I characteristics of SCR and measure latching and holding currents.
3. To compare the R, RC & UJT trigger circuit for SCR.
4. To study the commutation circuit for SCR.
5. To study single phase fully controlled bridge rectifiers with resistiveand inductive loads.
6. To study single phase fully controlled bridge rectifiers with DC motor load.
7. To study three-phase fully controlled bridge rectifier with resistive and inductive loads.
8. To study single-phase ac voltage regulator with resistive and inductive loads.
9. To study single phase cyclo-converter
10. To study the four quadrant operation of chopper circuit
11. To study MOSFET/IGBT based single-phase bridge inverter.

Software based experiments (Scilab/MATLAB or any equivalent open source software)

12. To obtain the simulation of single phase half wave controlled rectifier with R and RL load and plot load voltage and load current waveforms.
13. To obtain simulation of single phase fully controlled bridge rectifier and plot load voltage and load current waveform for inductive load.
14. To obtain simulation of single phase full wave ac voltage controller and draw load voltage and load current waveforms for inductive load.
15. To obtain simulation of step down dc chopper with L-C output filter for inductive load and determine steady-state values of output voltage ripples in output voltage and load current.

Text/Reference Books:

1. M.H. Rashid,"Power Electronics: Circuits, Devices & Applications", Pearson Education
2. D.W. Hart, "Introduction to Power Electronics" Prentice Hall Inc.

BCS653: COMPUTER NETWORKS LAB		
Course Outcome (CO):		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to:		
CO 1	Simulate different network topologies.	K_3, K_4
CO 2	Implement various framing methods of Data Link Layer.	K_3, K_4
CO 3	Implement various Error and flow control techniques.	K_3, K_4
CO 4	Implement network routing and addressing techniques.	K_3, K_4
CO 5	Implement transport and security mechanisms	K_3, K_4

DETAILED SYLLABUS

1. Implementation of Stop and Wait Protocol and Sliding Window Protocol.
2. Study of Socket Programming and Client – Server model
3. Write a code simulating ARP /RARP protocols.
4. Write a code simulating PING and TRACEROUTE commands
5. Create a socket for HTTP for web page upload and download.
6. Write a program to implement RPC (Remote Procedure Call)
7. Implementation of Subnetting .
8. Applications using TCP Sockets like a. Echo client and echo server b. Chat c. File Transfer
9. Applications using TCP and UDP Sockets like d. DNS e. SNMP f. File Transfer
10. Study of Network simulator (NS).and Simulation of Congestion Control Algorithms using NS
11. Perform a case study about the different routing algorithms to select the network path with its optimum and economical during data transfer. i. Link State routing ii. Flooding iii. Distance vector
12. To learn handling and configuration of networking hardware like RJ-45 connector, CAT-6 cable, crimping tool, etc.
13. Configuration of router, hub, switch etc. (using real devices or simulators)
14. Running and using services/commands like ping, traceroute, nslookup, arp, telnet, ftp, etc.
15. Network packet analysis using tools like Wireshark, tcpdump, etc.
16. Network simulation using tools like Cisco Packet Tracer, NetSim, OMNeT++, NS2, NS3, etc.
17. Socket programming using UDP and TCP (e.g., simple DNS, data & time client/server, echo client/server, iterative & concurrent servers)

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner It is also suggested that open source tools should be preferred to conduct the lab (C , C++ , Java , NS3, Mininet, Opnet, TCP Dump, Wireshark etc.

BCS063: Blockchain Architecture Design		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to		
CO 1	Describe the basic understanding of Blockchain architecture along with its primitive.	K_1, K_2
CO 2	Explain the requirements for basic protocol along with scalability aspects.	K_2, K_3
CO 3	Design and deploy the consensus process using frontend and backend.	K_3, K_4
CO 4	Apply Blockchain techniques for different use cases like Finance, Trade/Supply and Government activities.	K_4, K_5
DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction to Blockchain**: Digital Money to Distributed Ledgers, Design Primitives: Protocols, Security, Consensus, Permissions, Privacy. Blockchain Architecture and Design: Basic crypto primitives: Hash, Signature,) Hashchain to Blockchain, Basic consensus mechanisms	**08**
II	**Consensus:** Requirements for the consensus protocols, Proof of Work (PoW), Scalability aspects of Blockchain consensus protocols Permissioned Blockchains: Design goals, Consensus protocols for Permissioned Blockchains	**08**
III	**Hyperledger Fabric (A):** Decomposing the consensus process , Hyperledger fabric components, Chaincode Design and Implementation **Hyperledger Fabric (B):** Beyond Chaincode: fabric SDK and Front End (b) Hyperledger composer tool	**08**
IV	**Use case 1 :** Blockchain in Financial Software and Systems (FSS): (i) Settlements, (ii) KYC, (iii) Capital markets, (iv) Insurance **Use case 2:** Blockchain in trade/supply chain: (i) Provenance of goods, visibility, trade/supply chain finance, invoice management discounting, etc	**08**
V	**Use case 3**: Blockchain for Government: (i) Digital identity, land records and other kinds of record keeping between government entities, (ii) public distribution system social welfare systems Blockchain Cryptography, Privacy and Security on Blockchain	**08**

Text books:

1. Mstering Bitcoin: Unlocking Digital Cryptocurrencies, by Andreas Antonopoulos
2. Blockchain by Melanie Swa, O'Reilly
3. Hyperledger Fabric - https://www.hyperledger.org/projects/fabric
4. Zero to Blockchain - An IBM Redbooks course, by Bob Dill, Davi Smits -https://www.redbooks.ibm.com/Redbooks.nsf/RedbookAbstracts/crse0401.html

BCS064: Data Compression		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to		
CO 1	Describe the evolution and fundamental concepts of Data Compression and Coding Techniques.	K_1, K_2
CO 2	Apply and compare different static coding techniques (Huffman & Arithmetic coding) for text compression.	K_2, K_3
CO 3	Apply and compare different dynamic coding techniques (Dictionary Technique) for text compression.	K_2, K_3
CO 4	Evaluate the performance of predictive coding technique for Image Compression.	K_2, K_3
CO 5	Apply and compare different Quantization Techniques for Image Compression.	K_2, K_3
DETAILED SYLLABUS		**3-0-0**

Unit	**Topic**	**Proposed Lecture**
I	Compression Techniques: Loss less compression, Lossy Compression, Measures of performance, Modeling and coding, Mathematical Preliminaries for Lossless compression: A brief introduction to information theory, Models: Physical models, Probability models, Markov models, composite source model, Coding: uniquely decodable codes, Prefix codes.	**08**
II	The Huffman coding algorithm: Minimum variance Huffman codes, Adaptive Huffman coding: Update procedure, Encoding procedure, Decoding procedure. Golomb codes, Rice codes, Tunstall codes, Applications of Hoffman coding: Loss less image compression, Text compression, Audio Compression.	**08**
III	Coding a sequence, Generating a binary code, Comparison of Binary and Huffman coding, Applications: Bi-level image compression-The JBIG standard, JBIG2, Image compression. Dictionary Techniques: Introduction, Static Dictionary: Diagram Coding, Adaptive Dictionary. The LZ77 Approach, The LZ78 Approach, Applications: File Compression-UNIX compress, Image Compression: The Graphics Interchange Format (GIF), Compression over Modems: V.42 bits, Predictive Coding: Prediction with Partial match (ppm): The basic algorithm, The ESCAPE SYMBOL, length of context, The Exclusion Principle, The Burrows-Wheeler Transform: Moveto-front coding, CALIC, JPEG-LS, Multi-resolution Approaches, Facsimile Encoding, Dynamic Markoy Compression.	**08**
IV	Distortion criteria, Models, Scalar Ouantization: The Quantization problem, Uniform Quantizer, Adaptive Quantization, Non uniform Quantization.	**08**
V	Advantages of Vector Quantization over Scalar Quantization, The Linde-Buzo-Gray Algorithm, Tree structured Vector Quantizers. Structured VectorQuantizers.	**08**

Text books:

1. Khalid Sayood, Introduction to Data Compression, Morgan Kaufmann Publishers
2. Elements of Data Compression,Drozdek, Cengage Learning
3. Introduction to Data Compression, Second Edition, Khalid Sayood,The Morgan aufmann Series
4. Data Compression: The Complete Reference 4th Edition byDavid Salomon, Springer Text Compression1st Edition by Timothy C. Bell Prentice Hall

BEE061: SPECIAL ELECTRICAL MACHINES

Pre-requisites of course: Electrical Machines-I & Electrical Machines-II.

Course Outcomes:		**Knowledge Level, KL**
Upon the completion of the course, the student will be able to:		
CO 1	Describe the working principle, Constructional Features of different types of electrical machines including the fractional kilowatt machines.	**K2**
CO2	Analyse torque- speed characteristics of different electrical machines and interpret their performance and identify the suitable machine for an operation.	**K4**
CO3	Study different types of control techniques for a machine and identify the best control strategy based upon different constraints.	**K4**
CO4	Illustrate the use of stepper, BLDCs, SRM, and other special machines in the area of the various industrial and domestic as well as commercial applications of various fractional kilowatt machines.	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1 – Remember K_2 – Understand K_3 – Apply K_4 – Analyze K_5 – Evaluate K_6 – Create

Detailed Syllabus:

Unit-I: Induction Machines: Concept of constant torque and constant power controls, SEIG, DFIG: Operating Principle, Equivalent Circuit, Characteristics, Applications, Linear Induction Motors. Construction, principle of operation, Linear force, and applications.

Two Phase AC Servomotors: Construction, torque-speed characteristics, performance and applications.

Unit-II: Stepper Motors: Constructional features, Principle of operation, Variable reluctance motor, Hybrid motor, Single and multistack configurations, Torque equations, Characteristics, Drive circuits, Microprocessor control of stepper motors, Closed loop control, Applications.

Unit-III: Switched Reluctance Motors: Constructional features, Rotary and Linear SRM, Principle of operation, Torque production, performance characteristics, Methods of Rotor position sensing, Sensor less operation, Closed loop control and Applications

UNIT-IV Permanent Magnet Machines: Permanent Magnet synchronous generator Operating Principle, Equivalent Circuit, Characteristics, Permanent magnet DC motors, sinusoidal PMAC motors, their important features and applications, PCB motors,

Permanent Magnet Brushless D.C. Motors: Principle of operation, Types, Magnetic circuit analysis, EMF and torque equations, Commutation, Motor characteristics and control, Applications.

UNIT-V: Single phase synchronous motor; construction, operating principle and characteristics of reluctance and hysteresis motors;

Single Phase Commutator Motors: Construction, principle of operation, characteristics of universal and repulsion motors.

TEXT BOOKS:

1. K.Venkataratnam, 'Special Electrical Machines', Universities Press (India) Private Limited, 2008.
2. T.J.E Miller, 'Brushless Permanent Magnet and Reluctance Motor Drives', Clarendon Press, Oxford,

1989.
3. P.S. Bimbhra "Generalized Theory of Electrical Machines" Khanna Publishers.

Reference Books:

1. T. Kenjo, ‘Stepping Motors and Their Microprocessor Controls’, Clarendon Press London, 1984.
2. M.G. Say "Alternating current Machines" Pitman & Sons.

BEEZ061: Web Technology

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to:		
CO 1	Understand the fundamental concepts of web development, including the history, protocols, and tools. Apply HTML and XML in the development of web projects.	K_3, K_6
CO 2	Apply CSS for designing and styling web pages, including the use of CSS properties, styling elements, and advanced techniques for creating responsive web sites.	K_2, K_3
CO 3	Develop interactive web applications using JavaScript and AJAX, with a focus on scripting documents, forms, and networking concepts such as internet addressing and TCP/IP sockets.	K_3, K_6
CO 4	Design and implement server-side applications using Enterprise Java Beans (EJB) and Node.js, including the creation of JavaBeans, RESTful APIs, and database operations with MongoDB.	K_2, K_4, K_6
CO 5	Implement web server functionality using Servlets and Java Server Pages (JSP), focusing on handling HTTP requests, session tracking, and utilizing custom tag libraries for dynamic web content.	K_2, K_3, K_4
DETAILED SYLLABUS		**3-0-0**

Unit	Topic	Proposed Lecture
I	**Introduction:** Introduction and Web Development Strategies, History of Web and Internet, Protocols Governing Web, Writing Web Projects, Connecting to Internet, Introduction to Internet services and tools, Introduction to client-server computing. **Web Page Designing:** HTML: List, Table, Images, Frames, forms, XML: Document type definition (DTD), XML schemes, Object Models, presenting and using XML, Using XML Processors: DOM and SAX.	**08**
II	**CSS:** Creating Style Sheet, CSS Properties, CSS Styling (Background, Text Format, Controlling Fonts), Working with block elements and objects, Working with Lists and Tables, CSS Id and Class, Box Model (Introduction, Border properties, Padding Properties, Margin properties) **CSS Advanced** (Grouping, Dimension, Display, Positioning, Floating, Align, Pseudo class, Navigation Bar, Image Sprites, Attribute sector), CSS Color, Creating page Layout and Site Designs.	**08**
III	**Scripting:** Java script: Introduction, documents, forms, statements, functions, objects, introduction to AJAX. **Networking:** Internet Addressing, InetAddress, Factory Methods, Instance Methods, TCP/IP Client Sockets, URL, URL Connection, TCP/IP Server Sockets, Datagram.	**08**
IV	**Enterprise Java Bean:** Creating a JavaBeans, JavaBeans Properties, Types of beans, Stateful Session bean, Stateless Session bean, Entity bean. **Node.js:** Introduction, Environment Setup, REPL Terminal, NPM (Node Package Manager) Callbacks Concept, Events, Packaging, Express Framework, Restful API. **Node.js with MongoDB:** MongoDB Create Database, Create Collection, Insert, delete, update, join, sort, query.	**08**
V	**Servlets:** Servlet Overview and Architecture, Interface Servlet and the Servlet Life Cycle, Handling HTTP get Requests, Handling HTTP post Requests, Redirecting Requests to Other Resources, Session Tracking, Cookies, Session Tracking with Http Session **Java Server Pages (JSP):** Introduction, Java Server Pages Overview, A First Java Server Page Example, Implicit Objects, Scripting, Standard Actions, Directives, Custom Tag Libraries	**08**

Text books:

1. Burdman, Jessica, “Collaborative Web Development” Addison Wesley 2.Xavier, C, “Web Technology and Design” , New Age International
2. Ivan Bayross,” HTML, DHTML, Java Script, Perl & CGI”, BPB Publication
3. Hans Bergsten, “Java Server Pages”, SPD O’Reilly
4. Margaret Levine Young, “The Complete Reference Internet”, McGrawHill.
5. Greg Lim, “Beginning Node.js, Express & MongoDB Development”, 1 September 2020, Greg Lim
6. Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, “MongoDB: The Definitive Guide, 3rd Edition”, December 2019, O'Reilly Media, Inc.

BEEZ062: Theory of Automata and Formal Languages		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand:		
CO 1	Analyse and design finite automata, pushdown automata, Turing machines, formal languages, and grammars.	K_4, K_6
CO 2	Analyse and design, Turing machines, formal languages, and grammars	K_4, K_6
CO 3	Demonstrate the understanding of key notions, such as algorithm, computability, decidability, and complexity through problem solving	K_1, K_5
CO 4	Prove the basic results of the Theory of Computation.	K_2, K_3
CO 5	State and explain the relevance of the Church-Turing thesis.	K_1, K_5
DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Basic Concepts and Automata Theory:** Introduction to Theory of Computation-Automata, Computability and Complexity, Alphabet, Symbol, String, Formal Languages, Deterministic Finite Automaton (DFA)- Definition, Representation, Acceptability of a String and Language, Non Deterministic Finite Automaton (NFA), Equivalence of DFA and NFA, NFA with ε-Transition, Equivalence of NFA's with and without ε-Transition, Finite Automata with output- Moore Machine, Mealy Machine, Equivalence of Moore and Mealy Machine, Minimization of Finite Automata.	**08**
II	**Regular Expressions and Languages:** Regular Expressions, Transition Graph, Kleen's Theorem, Finite Automata and Regular Expression- Arden's theorem, Algebraic Method Using Arden's Theorem, Regular and Non-Regular Languages- Closure properties of Regular Languages, Pigeonhole Principle, Pumping Lemma, Application of Pumping Lemma, Decidability- Decision properties, Finite Automata and Regular Languages	**08**
III	**Regular and Non-Regular Grammars**: Context Free Grammar (CFG)-Definition, Derivations, Languages, Derivation Trees and Ambiguity, Regular Grammars-Right Linear and Left Linear grammars, Conversion of FA into CFG and Regular grammar into FA, Simplification of CFG, Normal Forms- Chomsky Normal Form(CNF), Greibach Normal Form (GNF), Chomsky Hierarchy, Programming problems based on the properties of CFGs.	**08**
IV	**Push Down Automata and Properties of Context Free Languages**: Nondeterministic Pushdown Automata (NPDA)- Definition, Moves, A Language Accepted by NPDA, Deterministic Pushdown Automata(DPDA) and Deterministic Context free Languages(DCFL), Pushdown Automata for Context Free Languages, Context Free grammars for Pushdown Automata, Two stack Pushdown Automata, Pumping Lemma for CFL, Closure properties of CFL, Decision Problems of CFL, Programming problems based on the properties of CFLs.	**08**
V	**Turing Machines and Recursive Function Theory**: Basic Turing Machine Model, Representation of Turing Machines, Language Acceptability of Turing Machines, Techniques for Turing Machine Construction, Modifications of Turing Machine, Turing Machine as Computer of Integer Functions, Universal Turing machine, Linear Bounded Automata, Church's Thesis, Recursive and Recursively Enumerable language, Halting Problem, Post's Correspondance Problem, Introduction to Recursive Function Theory.	**08**

Text books:

1. Introduction to Automata theory, Languages and Computation, J.E.Hopcraft, R.Motwani, and Ullman. 2nd edition, Pearson Education Asia
2. Introduction to languages and the theory of computation, J Martin, 3rd Edition, Tata McGraw Hill
3. Elements and Theory of Computation, C Papadimitrou and C. L. Lewis, PHI